Neural State Geometry and Dynamical Measurement Families Across Sleep

A support-audited single-recording development pilot

Article type: Research Article

Short title: Neural State Geometry Across Sleep

Keywords: sleep, neural state, MNPS, diffusion geometry, EEG, measurement

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Abstract We report a support-audited development analysis of global MNPS coordinates in one verified ds005555 sleep recording. Exact epoch export retained 1,068 windows; 1,059 had known stage labels and passed the specified coordinate/QC rules. We summarized all 3D and 9D coordinates by recorded label and applied a fixed sensitivity rule requiring matching labels in both adjacent windows. This retained 935 windows, including only two N1 windows and no N3 windows. Primary coordinate medians varied across labels, but the loss of N1/N3 support limits interpretation. Full filter support and neighborhood stage purity remain unverified, and the source includes a non-equivalent PPG fallback. These results establish a reproducible descriptive workflow and identify support limitations; they do not establish population discrimination, a consciousness marker, or stage-conditioned dynamics.

Author Summary

A single-recording sleep pilot illustrates why measurement availability and temporal support must accompany neural state summaries. A fixed label-boundary check removes all N3 windows and nearly all N1 windows, preventing strong conclusions about those stages from this recording.

1. Introduction

Sleep recordings provide repeated observations of changing neural states, but temporal gaps, window support, and repeated-measure structure can make a computed endpoint unsuitable for a stage comparison. This pilot therefore treats support and measurement status as part of the result.

Core question

Can the recorded MNPS and dynamical measurement families be aligned to sleep stages with explicit temporal support and reproducible validity metadata?

2. Theory and measurement contract

MNPS coordinates describe a state representation used by the NeuralManifoldDynamics pipeline. Diffusion geometry is considered when its canonical family status, coordinate contract, and support are valid. A transition covariance proxy and its propagated W_Q summaries retain recording grain; they are not treated as stage-native or individual-level noise measurements.

The 9D volume statistic `v_norm` is currently method-limited: the pilot shows sensitivity, but does not isolate conditioning, projection, and eigenvalue-arithmetic contributions. It remains outside biological interpretation pending an independent numerical reference.

3. Methods

3.1. Dataset and staged support

The source is the verified ds005555 sub-99 recording. The read-only epoch export retains source, configuration, code, and output hashes. Recorded window labels are aligned to MNPS coordinates using exact source IDs and time bounds. These labels do not establish containment of the full filter support or diffusion neighborhood within one stage. Any future stage-conditioned endpoint requires a separate full-support assessment.

A bounded annotation audit compares the epoch table with the authoritative human-scored `stage_hum` source. It verifies row-to-epoch mapping, onset and 30-second duration, and the 256-Hz sample clock from the pinned EEG metadata. Code 8 is retained as an unresolved PSG-disconnection status; `stage_ai` is not used as a replacement. The raw recording lasts 32,064 seconds while the accepted epoch bounds end at 32,040 seconds, leaving a 24-second tail outside the epoch table. These checks establish annotation and table alignment only; full filter support and diffusion-neighborhood purity remain unassessed.

3.2. Descriptive development experiment

We summarize the recorded global coordinate representation by window label, without estimating separate stage-conditioned geometries. Eligibility is evaluated separately for 3D and 9D: a known stage, a finite complete vector, true coverage and retained-after-QC masks, and the corresponding producer geometry-valid or 9D-coordinate-valid mask. Missing masks cannot pass. Counts and coordinate medians with 25th and 75th percentiles are retained for every stage under both analysis policies.

The fixed label-boundary sensitivity subset additionally requires the preceding and following windows to share the same known label, with consecutive IDs and touching time bounds. Endpoints, gaps and unknown labels cannot pass this rule. It is a three-window label diagnostic, not proof of full temporal support or neighborhood purity. The policy was written after the source support audit, before inspecting the new descriptive outputs; it is not a preregistration.

The source uses a PPG-rate fallback for `e_m`, with `semantic_equivalence=false`, and the composite e-axis has mixed source provenance. This prevents interpreting the representation as an exclusively EEG-derived layer. Complete anchor identifiers and full filter-support metadata have not been verified.

3.3. Primary endpoints

A separate candidate-neighborhood audit reuses the stored 3D coordinates, timestamps, source-file segmentation and pinned diffusion settings ($k = 20$). Candidate membership is exported only after exact agreement with stored tensor values at their stored dtype, support counts and validity, followed by an exact selected-increment covariance check. Source snapshots and before/after hashes are retained. Diffusion is window-native with a recording parent; its global neighbor pool is not restricted by stage. Matching tensors do not prove unique historical neighbor identities, which were not persisted in the original HDF5.

The primary endpoint family is MNPS state geometry, followed by diffusion-family summaries where support and status permit. Q/W_Q, destination, resilience, and DF-DRIFT-C1 are recorded with their native status and grain; unavailable or method-limited outputs remain unavailable or descriptive.

3.4. Statistical design

The first pass is descriptive and within participant/recording. Any later cohort analysis will freeze participant inclusion, aggregation, contrast, effect size, and correction family after the support audit and before reading contrasts. Windows and transitions are repeated measurements and are never counted as independent participants.

3.5. Robustness and missingness

The audit reports temporal gaps, irregular sampling, insufficient support, nonfinite values, coordinate contracts, computation status, measurement validity, and missingness by stage and endpoint. No refusal is converted to zero. All preselected diagnostic horizons remain visible.

4. Results

4.1. Verified export and coverage

The accepted epoch export contains 1,068 non-overlapping 30-second windows spanning 32,040 seconds. Nine labels are unknown. All coordinate vectors are finite and all four recorded producer masks are true, giving 1,059 eligible windows in each representation after excluding unknown labels. Re-running the integrated exporter produced a byte-identical Parquet table. Independent recomputation reproduced all 120 coordinate-by-stage summary rows across the two representations and two policies.

The run-support audit identifies 84 contiguous label/status bouts: 83 known-stage bouts and one unresolved code-8 disconnection bout. Known-stage counts are Wake 17 bouts/275 windows, N1 16/20, N2 33/642, N3 15/15, and REM 2/107. All 15 N3 bouts are single 30-second windows, which accounts for the loss of N3 under the fixed two-sided boundary diagnostic. This is a support limitation in one recording, not evidence of annotation error or absent N3 physiology.

Stage	Primary n	Boundary n	m median	d median	e median
Wake	275	249	0.922	1.769	0.336
N1	20	2	0.679	1.354	0.132
N2	642	581	−0.344	−0.255	−0.121
N3	15	0	−1.392	−1.007	−0.255
REM	107	103	0.701	0.190	0.121

Table 1: Recorded-label counts and primary 3D medians. Counts are windows from one recording, not participants. The boundary rule requires both immediate labels to match. Unknown labels (n = 9) remain in the audit. Full quartiles and both-policy summaries for all 12 coordinates are retained in the accepted run.

4.2. Coordinate summaries and label-boundary sensitivity

Primary m and d medians differ descriptively among the recorded labels (Table 1). The sensitivity subset retains 935 windows: 249 Wake, 2 N1, 581 N2, no N3, and 103 REM. N3 sensitivity statistics therefore remain unavailable, rather than zero. The two remaining N1 windows do not provide a stable basis for a stage summary.

For transparency, boundary-subset medians (m, d, e) are (1.019, 1.775, 0.333) for Wake, (-0.334, −0.253, −0.116) for N2, and (0.702, 0.189, 0.131) for REM. N1 values are retained in the complete tables with n = 2. These are window distributions in one global representation; no population test, classification performance, or separate stage-conditioned estimator is reported.

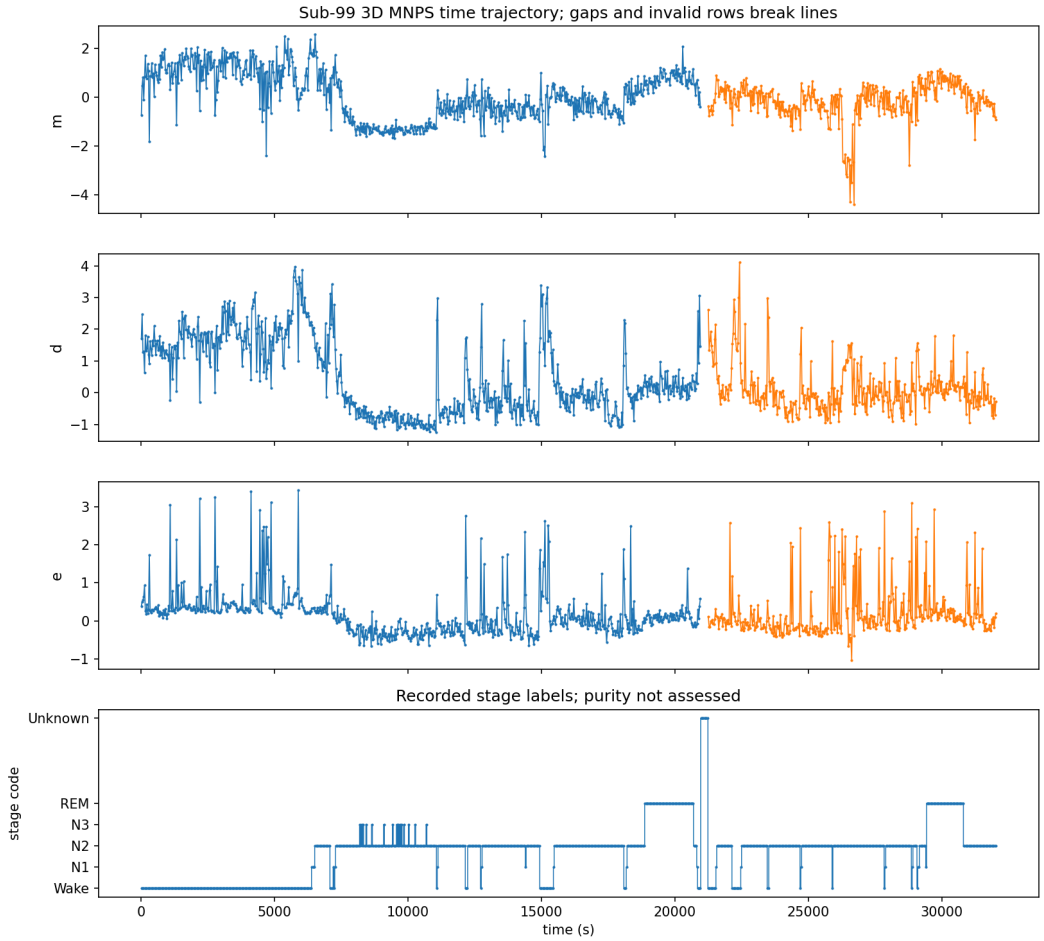


Figure 1: Global MNPS coordinates over time and recorded sleep labels in sub-99. Unknown-label windows are omitted from coordinate traces and retained in the label panel. Trace colors distinguish contiguous plotted segments, not stages. The composite e-axis includes mixed provenance and the 9D e_m source is a non-equivalent PPG fallback. Aligned labels do not establish full filter support or stage purity.

4.3. Dynamical-family availability

The earlier accepted import audit retained 1,068 computed diffusion tensor rows, with finite D_total, d_diff and c_diff. Drift-dependent A_bD and R_b_over_a remain unavailable because drift was not supplied. These are

availability findings from the import audit; the present experiment does not add sleep-stage diffusion contrasts. Q/W_Q retains recording grain, and the method-limited 9D volume endpoint remains outside biological interpretation.

4.4. Candidate diffusion support

The candidate reconstruction matched all 1,068 stored tensors and exported 21,360 selected-increment rows, with no exact-distance cutoff ties detected. For 281 of 1,059 known-stage queries (26.5%), all 20 selected increments have both endpoints in the query's stage. Across all queries, 88 selected-increment rows touch a code-8 disconnection endpoint; 33 known-stage queries include at least one such increment. These are repeated selections, not independent events or participants. An independent direct calculation reproduced every selection and tensor without importing the producer or reconstruction runner. This establishes consistency of the candidate reconstruction with the stored global estimator. It does not establish historical membership, complete feature/filter support, or separately estimated stage-conditioned diffusion.

5. Discussion

The subsequent fixed increment-support audit separates structural validity, same-known-stage endpoint labels, and full signal support. All 1,067 adjacent pairs satisfy the structural rule, including the stored 30-second nominal dt and 0.05 relative-deviation limit. Of these, 976 have matching known-stage endpoints: 258 Wake, 4 N1, 609 N2, 0 N3 and 105 REM. Ten adjacent pairs touch a code-8 disconnection. These pool counts differ from repeated candidate neighbor selections. Full-support-qualified counts remain unavailable, not zero; no tensor was refitted from the restricted pool.

Feature-support inspection identifies a recording-global robust-z reference fit over 1,068 rows. The 30-second feature input intervals do not establish the effective support of preceding filters or resampling. Executed filter lengths, padding and complete branch provenance are not sufficiently recorded. The embedded time-grid audit resolves the generic 8/4 settings to the actual 30/30-second epoch profile; this distinction is not a detected time-axis error. Reference-fit dependence and raw temporal support must remain separate.

These results describe one global representation and dependent windows from one recording. They cannot isolate sleep-stage effects from time of night, preprocessing or mixed feature provenance. The absence of supported N3 increments is not evidence of absent N3 physiology or incorrect annotations.

The next gate is an executed signal-support export and a bounded reference check, followed by a frozen analysis and participant-level replication protocol. The local participant table has 128 rows and 100 distinct pid values, whereas the README reports 108 people; this discrepancy remains unresolved. Q/W_Q and method-limited 9D volume remain outside biological conclusions. No consciousness-theory or clinical-biomarker validation is established.

6. Data and code availability

Accepted article artifacts are descriptive r10, stage audit r7, diffusion candidate r9, and increment-support audit `increment_support_sub99_20260908_r2`. Their results directories retain source, code, configuration and output hashes, complete tables, independent reviews and coordinator recomputation evidence. Earlier development candidates are preserved. Raw HDF5 files remain outside the article directory. Public source citation and submission metadata still require verification before submission.

Author Contributions

RL: Conceptualization, Methodology, Data Curation, Analysis, Writing, and Editing.

Funding Information

Funding information will be reported after source and grant records are verified.

Data Availability Statement

The source dataset and analysis artifacts will be identified by their public source records, local manifests, and cryptographic hashes. Raw HDF5 files are not stored in this article directory.

Competing Interests

The author declares no competing interests, subject to final confirmation at submission.